

White paper: the pmsimstats-ng manuscript compendium

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A five-page review of the eleven manuscripts under `analysis/report/`, with an assessment of the contribution each makes and a prioritized set of next steps.

Scope and method of this review

The compendium comprises eleven manuscripts, each in its own numbered directory, all targeting *Statistics in Medicine* style and sharing a common preamble, bibliography apparatus, and simulation substrate. Several directories carry `report-slim`, `report-extra-slim`, `report-mini`, or `report_short` variants; these are length-reduced renderings of the same study, not separate investigations, and are assessed here as one paper each.

This review is source-level. Every `report.Rmd` was read for its abstract, section structure, and results, together with the project READMEs, the six committed referee reports, the paper-04 remediation plan, and the three existing per-paper white papers (01, 02, 03). No simulation was re-executed and no numerical result was independently recomputed for this document, so all quantitative statements below are reported as the manuscripts report them. Assessments of contribution and of weakness are the reviewer’s judgment; claims about draft state (placeholder counts, missing results, deferred sweeps) are verified by inspection of the sources.

What the programme is

The compendium is a single coherent research programme rather than a collection of related papers. Its object is one estimand: the biomarker-by-treatment interaction in aggregated N-of-1 trials, the quantity such trials exist to estimate when they are used to validate a predictive biomarker. Its motivating application throughout is the prazosin/PTSD nightmare-frequency setting of Hendrickson et al. (2020), and its method throughout is Monte Carlo simulation reported under the ADEMP framework of Morris, White, and Crowther (2019).

The eleven papers partition into five layers.

- **Generative substrate.** 06 (three-component BR/PB/TV decomposition) and 07 (Gompertz trajectory template) justify the shape of the simulated data.
- **Interaction encoding.** 01 (Architecture A vs B), 11 (Architecture C, combined), and 03 (latent-class and mixture formulations) ask how the biomarker’s moderating effect should enter the data-generating process at all.
- **Analyst-side inference.** 02 (carryover specification A1/A2/A3), 08 (RM-ANOVA vs mixed model vs GEE), and 10 (working-covariance miscalibration and cluster-robust repair) ask what the analyst should fit.
- **Design.** 05 (twelve-sweep design sensitivity), 09 (informative dropout by design family), and the design axis of 08.
- **Clinical framing.** 04 (N-of-1 versus parallel-group RCT for the treatment main effect).

The programme’s intellectual center is paper 01. Its finding, that the carryover penalty is an order of magnitude larger when the interaction is encoded in the covariance (Architecture B) than in the mean (Architecture A), propagates as a conditioning variable into 02, 03, 07, and 11, and it is the reason those papers report architecture-stratified rather than pooled conclusions. That is a genuine and unusual structural virtue: the programme has identified a modeling choice that the surrounding literature treats as an implementation detail and has made it an explicit axis everywhere downstream.

Paper-by-paper assessment

01. Two (three) data-generating architectures. *Useful; the keystone paper.* Establishes that Architecture B loses 30.6% relative power at a one-week carryover half-life in the OL+BDC design against 2.8% for Architecture A, with a difference-of-differences $z = 7.64$, while the crossover design shows no architecture effect. The mechanism is argued cleanly: Architecture B’s signal lives in a correlation that decays off drug, so carryover attacks it twice. Type I error is checked across 18 null cells and an $N = 70$ robustness run preserves the ordering. This paper should be submitted first; the rest of the compendium cites it. Two weaknesses persist. The Architecture C material now duplicated in paper 11 uses an additive-on-the-probability-scale benchmark to claim super-additivity, which is the wrong benchmark for a convex power function and inflates the apparent effect from roughly three Monte

Carlo standard errors to nine. Section 6’s power-recovery strategies are qualitative hypotheses with no simulation behind them.

02. Carryover-mitigation analysis strategy. *Useful; the strongest negative result in the compendium.* A 540-cell factorial establishes that the exposure-weighted specification A2, the framework’s own default, attains the highest power in only 19% of non-null cells. It wins by about ten points under Architecture B in designs with a blinded discontinuation phase and loses badly under the classical crossover (0.488 against 0.830). A3, the Jones-Kenward lagged-term device, is shown to be numerically almost identical to A1 because its lagged term enters only as a main-effect nuisance. The paper also reports that 30% dropout collapses all three specifications to indistinguishability, which is the most actionable single sentence in the compendium. Remaining weaknesses: bias, MSE, and coverage in the headline table are scored against A2’s estimand and are therefore not comparable across specifications (the paper says so, but the table invites the comparison anyway); the CR2 recalibration is validated in one stratum only; and the previously noted inconsistency with paper 01’s carryover-loss figures should be re-verified after 01’s final numbers settle.

03. Latent-class and mixture formulations. *Partially useful; a theory paper with a pilot attached.* Aims 1 and 2 (formalizing the latent-class DGP, deriving when a single MVN substitutes for it, characterizing identifiability) are delivered and are the paper’s real contribution, particularly the result that the achievable marginal correlation is bounded by the geometric mean of the two class separations. Aim 3 is a 240-replicate pilot of the last of five pre-registered studies; aim 4, the prazosin re-analysis, has not been attempted. The pilot’s findings are sharp (the unconditional class-aware Wald test rejects at 0.197 against nominal 0.05; the BIC-conditional test controls size but is essentially inert at 0.025 power), but they rest on one sample size, one design, and one generative mechanism, and the regime where a class-aware test should win, large separation with genuinely bimodal response, was never tested. As drafted the paper claims less than it appears to; it should either be reframed explicitly as theory-plus-pilot or the canonical 1,000-replicate run should be executed before submission.

04. N-of-1 versus parallel-group RCT. *Useful clinically, but carries an unaddressed comparison-fairness problem.* Reports that the aggregated N-of-1 hybrid dominates the parallel RCT across the entire effect-size range, with roughly a threefold power ratio at the moderate-effect cell, negligible bias, and near-nominal coverage in both arms. The ADEMP apparatus is the most complete in the compendium and the 2026-05-19 remediation plan’s critical items (dual implementations, RNG discipline) appear to have been carried out; the DerSimonian-Laird estimator now survives only as literature discussion. The problem is the comparison itself. Both arms are matched at $N = 35$ participants, but the N-of-1 arm contributes eight periods per participant against the RCT’s single post-baseline measurement, so the result substantially restates that more measurements per subject buy more information. The Limitations section covers compliance, single outcome, and decay form, but does not mention observation count or participant burden at all. A referee at a general medical venue will raise this immediately. Either add a measurement-matched or burden-matched sensitivity arm, or state the asymmetry explicitly and defend it as the decision-relevant comparison for a fixed recruitable population.

05. Twelve-sweep design sensitivity. *Useful as a reference appendix; weak as a standalone paper.* The twelve sweeps (S1-S12) are well chosen, each anchored to a literature precedent, and the null calibration sweep at 2,000 replicates is good practice. The synthesis is defensible: the Hendrickson hybrid is a competent default; ordering flips to the parallel RCT only when carryover exceeds the within-subject period length. But the paper is organized as twelve one-factor-at-a-time sweeps with a single factorial (S11), so it characterizes the axes rather than their interactions, and its target estimand is the main effect, which makes it a companion to 04 rather than to the interaction papers that constitute the programme’s core. Its most likely fate is as supplementary material to 04.

06. Three-component decomposition. *Useful; the most intellectually substantial paper after 01.* The omitted-variable-bias identity is the contribution: the one-component biomarker slope is displaced from the pharmacological slope by a term proportional to the biomarker’s covariance with the placebo-belief and natural-history components, not by the magnitude of those components. Study A confirms that an uncontaminated biomarker suffers essentially no bias and that the cheap phase-augmented remedy is strictly dominated; the contaminated-biomarker pilot exhibits the failure mode at +0.51 bias; Study B shows recovery is possible only under a balanced-placebo design that decouples exposure from belief. That last result, that the cure is a design property rather than an analysis property, is the paper’s real payload and is what the fourth referee report was asking for. Weaknesses: the paper is 26,000 words of source, by a wide margin the longest in the compendium, and needs the slim variant promoted to canonical; the contaminated-biomarker result rests on a 100-replicate pilot while the uncontaminated case gets 1,000; and the practical recommendation (balanced-placebo designs) is rarely feasible in the clinical settings the programme targets, which the Discussion should concede more squarely.

07. Gompertz trajectory evaluation. *Useful, and materially improved by revision.* The first-round referee identified a fatal construction flaw, that the four trajectory families were not four data-generating processes but a post-hoc imposition on Gompertz-generated data. The current draft reports an architecture-dependent answer instead: under Architecture B the trajectory family is exactly inert because the interaction is mean-independent, so matched seeds return identical rejection decisions; under a faithfully multiplicative Architecture A the four families separate by 0.039 in power with Gompertz the mildly conservative member. This is a better paper than the one reviewed, and the exact-inertness result is a clean structural finding worth stating on its own. It remains a narrow paper: one design (OL+BDC), one sample size, zero carryover, 16 cells. The regimes the README promises to identify (non-saturating growth, biphasic patterns, breakpoint behavior) are discussed but not simulated, and the zero-carryover choice is awkward in a programme whose central finding is about carryover.

08. Test procedure and design. *Useful; the cleanest practical message in the compendium.* The closed-form RM-ANOVA derivation is correct per the referee’s independent re-derivation, and the revised simulation now separates the two effects that the first draft confounded: dichotomizing the biomarker costs 0.08 to 0.21 in power, while the further step from a continuous-biomarker contrast to the full mixed model adds at most 0.04. Naive GEE inflates Type I error to 1.5-1.9 times nominal at all four sample sizes and the Mancl-DeRouen correction repairs it at roughly ten points of power. The dichotomization result is the finding most likely to change applied practice, and it deserves to be the paper’s headline rather than a component of it. The unresolved problem is scope: the cycle-by-period design grid is specified in detail and then deferred, which leaves the title’s second half unearned. Either run the grid or retitle the paper around the test-procedure axis.

09. Informative dropout by design. *Promising but not yet a paper.* The gap is real and correctly identified: Hendrickson’s Figure 4A implies a design-by-dropout coupling that the N-of-1 methods literature has not pursued, and no other manuscript here centers dropout. But the abstract’s Results paragraph still reads “[Forthcoming]”, the empirical content is a 500-replicate 16-cell prototype at one sample size, one value of c_{bm} , and Architecture A only, and the same prototype paragraph appears verbatim in both Results and Discussion. More seriously, the reported power ordering is measured at the ceiling: hybrid at approximately 1.00 and OL+BDC at approximately 0.95 cannot discriminate designs, and OL sits at nominal alpha by construction because it supplies no off-drug contrast. The design ordering claim therefore has almost no resolution in the cells that produced it. The “happy accident” randomization-path decomposition, which is the paper’s most original idea, needs a cell where power is near 0.5 to be visible at all.

10. Interaction-test calibration. *Useful, and the most portable paper in the compendium.* It is the only manuscript whose contribution is not specific to N-of-1 trials: the finding is that a misspecified parametric working covariance biases the model-based standard error of a mixed-model interaction asymptotically, in either direction. The diagnostic staging is convincing, ruling out degrees-of-freedom rules, positive-definiteness correction, and point-estimate bias, then localizing the 6-10% standard-error inflation to the corCAR1 residual layer, demonstrating that the bias does not attenuate from $N = 35$ to $N = 280$, and repairing calibration with a CR2 cluster-robust standard error and independently from the marginal side with bias-corrected GEE. The reusable staged protocol is a real deliverable. Two gaps: the worked example is the same prazosin cell as everything else, so the generality claim rests on argument rather than on a second application; and the paper has no README and no cover letter, indicating it has not been through the same submission-preparation pass as its neighbors.

11. Combined DGP architecture (Architecture C). *Not yet assessable; currently an extraction in progress.* The manuscript is 3,762 words of source against a compendium median of about 13,000, has no rendered PDF, and its abstract states “[To be finalized after the common-driver boundary re-run]”. Its content is the Architecture C material currently also present in paper 01, being split out per `01-.../architecture-c-extraction-plan.md`. Whether the split is warranted is doubtful: C is a sensitivity device rather than a co-equal third architecture, and the case for it is better made inside 01 than as an independent manuscript (see the consolidation section below). Two substantive issues must be fixed either way. The super-additivity claim needs the effect-size-scale benchmark rather than the probability-scale one, under which the observed excess shrinks to roughly three Monte Carlo standard errors. And the grid’s carryover comparison is currently reported only for the crossover design, the one design in which paper 01 found no architecture-by-carryover interaction; the informative panel is OL+BDC. The paper should also state plainly that $c_{bm,A}$ and $c_{bm,B}$ are not separately identifiable from a fitted `bm:Dbc` coefficient, which makes C a simulation-side construct only.

What the compendium establishes collectively

Read together, the eleven papers deliver a coherent and defensible set of claims about aggregated N-of-1 biomarker-validation trials.

1. How the interaction is encoded in the data-generating process is a scientific assumption with large consequences under carryover, not an implementation detail (01, with 03 and 11 mapping the neighborhood).
2. Analyst-side defaults are conditional, not universal. The exposure-weighted carryover predictor helps only under one architecture and one class of design (02); the mixed model beats GEE and RM-ANOVA, but most of that gap is dichotomization rather than the test (08); the model-based standard error is not trustworthy when the working covariance is a convenient parametric guess (10).
3. Design dominates analysis. Retention beats specification choice (02); the balanced-placebo contrast, not any estimator, is what recovers an unbiased pharmacological slope under a contaminated biomarker (06); design family governs dropout robustness (09).
4. Two modeling choices that might have been feared are largely inert: the trajectory family (07, exactly inert under Architecture B) and mixture versus continuous encoding at trial-relevant N (03, where the standard mixed model dominates a class-aware test).

Negative and inertness results of this kind are the compendium’s comparative advantage, and it is worth being deliberate about that in the cover letters. The programme has repeatedly tested its own defaults and reported when they failed.

Cross-cutting weaknesses

Author prose is absent everywhere. All eleven manuscripts use the three-part bullets / rgt / orig scaffold, and no rgt block in any paper contains author text: papers 01, 02, and 11 hold a “to complete” placeholder and the remaining eight hold Lorem ipsum, 645 blocks in total. This is the single binding constraint on submission and it is not a formatting task; the rgt layer is where the author’s voice and argument are supposed to live.

One reference parameter set. Nearly every simulation is calibrated to the same prazosin/PTSD trajectory. Absolute power values are therefore illustrative throughout, and only orderings port. Paper 10 is the case where this matters most, because its claim is explicitly general.

Ceiling saturation. Papers 09 and, in places, 04 and 05 report power at or near 1.00, where designs and methods cannot be discriminated. Cells should be chosen to sit near 0.5 for the contrast of interest.

Prototype-versus-production gap. Papers 03, 06 (contaminated pilot), 08 (deferred design grid), and 09 present pilots or partial grids as if they carried the weight of the pre-registered production runs. Each states this, but the abstracts do not always reflect it.

Duplicate and stale scaffolding. Several READMEs still assert that driver directories “do not yet exist” when they do; paper 09 repeats a paragraph verbatim across two sections; papers 10 and 11 lack READMEs and cover letters. These are cheap to fix and they signal draft state to a referee.

Recommended consolidation

Eleven manuscripts on a single estimand, calibrated throughout to one prazosin/PTSD parameter set, invites the charge of over-partitioning, and a referee who encounters two of them will notice. Six papers carry a defensible independent claim: 01 (the architecture distinction), 02 (the conditional failure of the exposure-weighted default), 06 (the omitted-variable identity and the design-side cure), 08 (the dichotomization cost), 10 (working-covariance miscalibration, the only contribution not specific to N-of-1 trials), and 04 (the clinical comparison, conditional on repairing its comparison basis). Three do not, and the objection in each case is to the packaging rather than the content, all of which is worth keeping.

- **Fold 11 back into 01** as a scoped sensitivity section rather than completing the extraction. Architecture C is the common generalization of A and B, not a rival to them, and its two channel parameters are not separately identifiable from a fitted bm:D_{bc} coefficient, so it is a simulation-side construct with no estimation counterpart. Its two legitimate uses, a boundary-cell correctness gate on the DGP implementation and a mixing-weight sweep when the mechanism is unknown, are arguments about how to use 01’s framework and read naturally as part of 01.
- **Publish 05 as supplementary material to 04.** Its twelve sweeps target the treatment main effect, not the interaction, which makes it a companion to 04 rather than to the programme’s core, and its one-factor-at-a-time structure characterizes axes rather than their interactions.
- **Merge 07 into 06 as a robustness section on the generative substrate.** Both concern how the simulated trajectories are shaped; neither is large on its own; and 07’s cleanest result, that trajectory family is exactly inert under a covariance-channel DGP, is a statement about the substrate that 06 formalizes.

If 06 is slimmed for submission as recommended above, 07's sixteen cells fit comfortably in the technical companion.

The result is seven or eight submittable papers of more even weight: 01 (absorbing 11), 02, 04 (with 05 appended), 06 (absorbing 07), 08, 10, and 03 and 09 once their outstanding simulation work completes.

Recommended next steps, in priority order

1. **Write the rgt prose for paper 01, then 02, then 10.** Nothing can be submitted until this is done, and 01 gates the citation chain for the rest. Papers 02 and 10 are the next most nearly complete. Run `~/bin/strip-claudecode` on each as its `rgt` layer closes.
2. **Settle the Architecture C question and re-verify 01's boundary cells.** Adopt the consolidation above (fold 11 into 01) or defend the extraction explicitly. Either way, recompute the super-additivity comparison on the effect-size scale, replace the crossover carryover panel with OL+BDC, and confirm that $(c_{bm,A} = 0.45, c_{bm,B} = 0)$ and $(0, 0.45)$ reproduce 01's published power figures exactly before accepting the grid. Re-check the 01/02 carryover-loss consistency afterward.
3. **Fix paper 04's comparison basis.** Add either a measurement-matched or a burden-matched RCT arm, or a stated, defended asymmetry, and put observation count in the Limitations. This is a one-sweep change and it removes the paper's most likely rejection reason.
4. **Re-scope paper 09 around resolvable cells.** Re-run the design-by-dropout grid at parameter values placing power near 0.5, add Architecture B, extend beyond a single c_{bm} , and write the Results paragraph of the abstract. Drop the duplicated prototype paragraph.
5. **Decide paper 08's scope.** Either run the cycle-by-period grid or retitle around the test-procedure axis and promote the dichotomization result to the headline.
6. **Promote slim variants to canonical for 06 and 02.** Both exceed plausible journal budgets in their full form. The full versions become technical companions in `docs/`.
7. **Add a second application to paper 10.** A non-N-of-1 longitudinal example, even a small one, converts its generality claim from argument to demonstration and makes the paper submittable to a broader methods venue.
8. **Run the canonical replication for paper 03, or reframe it.** A 1,000-replicate run at $N \in \{35, 70, 100\}$ over a wider gating-slope grid, including a genuinely bimodal separation regime, is required before the non-competitiveness result generalizes. If that compute is not available, reframe as theory plus pilot and say so in the abstract.
9. **Housekeeping.** Refresh the stale READMEs, add READMEs and cover letters for 10 and 11, update the compendium table in `analysis/report/README.md` to cover papers 10 and 11, and record the intended submission order.

A reasonable submission sequence, assuming the consolidation above, is 01 (absorbing 11) and 10 first, being the keystone and the most portable; 02 and 08 next, both close to complete and both carrying clean practical messages; then 06 in slim form with 07 as a robustness section; then 04 with 05 as supplementary material, once the comparison basis is repaired; with 03 and 09 last, as their outstanding simulation work completes.

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